# APPENDIX

## A. Comparison with Active Object Tracking Methods

Active Object Tracking (AOT) methods, as described in [29], utilize an end-to-end approach through reinforcement learning. These methods process raw video frames as input and generate camera movement actions as output. According to [29], there are seven discrete actions: *move-forward/backward, turn-left/right, move-forward-and-turn-left/right*, and *no-op*. However, existing person-following datasets, such as Chen's dataset [5] and our own dataset, only provide ground truth bounding boxes of the target person. For a valid comparison, we need to map these bounding boxes to the action space. As noted in [29], a two-stage method (combining single object tracking with a PID controller) can achieve a 100% success rate if the estimated target bounding box (bbox) is accurate. Therefore, we can deduce the ground truth actions from the ground truth target bboxes. This approach allows us to evaluate AOT methods within person-following datasets. According to [29], the objective of the camera action is to keep the target person's bbox centered in the image, maintaining the same size as initially observed. The ground truth action is generated according to the horizontal error $X_{err} = \frac{X_b - W/2}{W/2}$ and the size error $S_{err} = \frac{W_b \times H_b - W_{exp} \times H_{exp}}{W_{exp} \times H_{exp}}$ shown as in Fig. 5. According to [29], we have the following ground truth action mappings:

(1) *Move forward* if $\mathrm{abs}(X_{err}) \leq 0.1$ and $S_{err} \leq -0.2$;
(2) *Move backward* if $\mathrm{abs}(X_{err}) \leq 0.1$ and $S_{err} \geqslant 0.2$;
(3) *No-op* if $\mathrm{abs}(X_{err}) < 0.1$ and $\mathrm{abs}(S_{err}) < 0.2$;
(4) *Move forward and turn right* if $0.1 \leq X_{err} \leq 0.3$;
(5) *Turn right* if $X_{err} > 0.3$;
(6) *Move forward and turn left* if $-0.3 \leq X_{err} \leq -0.1$;
(7) *Turn left* if $X_{err} < -0.3$.

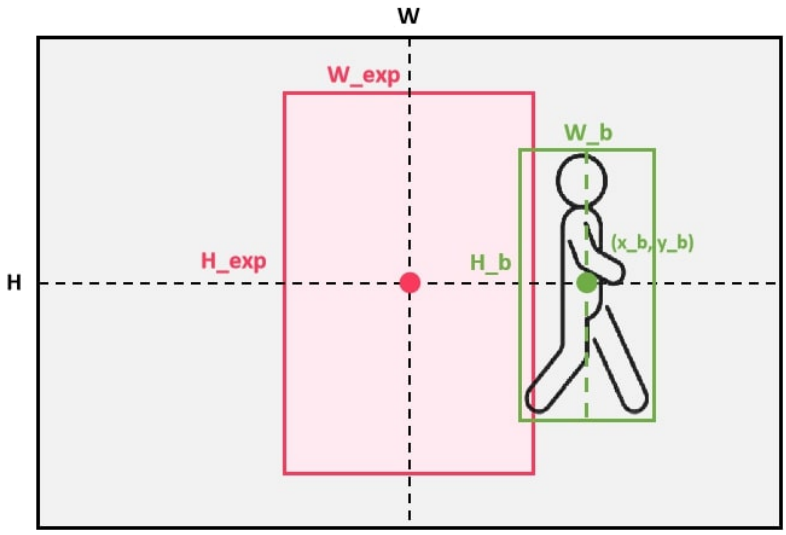



Fig. 5. An example to illustrate errors is as follows: The goal of a correct action is to move the target person closer to the center of the image. Here, $(W, H)$ represents the image's width and height. $(W_{exp}, H_{exp})$ denotes the size of the expected centered bounding box, initialized by the first bounding box of the target person. $(W_b, H_b)$ represents the target person's bounding box size in the current frame, and $(x_b, y_b)$ is its center point.

This paper aims to evaluate the algorithms' tracking performance, meaning true target person identification is the first priority. Therefore, besides evaluating accurate action estimation, we reduce the matching standards. Specifically, we consider an action to be true if this action tries to move the target person to the center of the image. For example, if the bbox of the target person is on the left of the image, *move backward, turn left*, and *move forward and turn left* are all considered as true actions. The results are reported in Table. I. An example is shown in Fig. 6. We observe that after a long occlusion by a visually similar person, Zhong's method outputs *move forward and turn left*, failing to re-identify the target person. In contrast, our method reliably re-identifies the target person even after long-term occlusion.

(a) Zhong's method [29]

(b) Ours

Fig. 6. An example comparing Zhong's method [29] and ours. From left to right, the sequence represents observations where a long-term occlusion occurs. (a) Zhong's method outputs *move forward and turn left* due to a failed re-identification of the target person. (b) Our method reliably re-identifies the target person even after long-term occlusion.

### B. Memory Examples

Several replayed examples of our short-term and long-term memories are shown in Fig. 7. For more visual examples of short-term and long-term memories, please refer to the supplementary video.

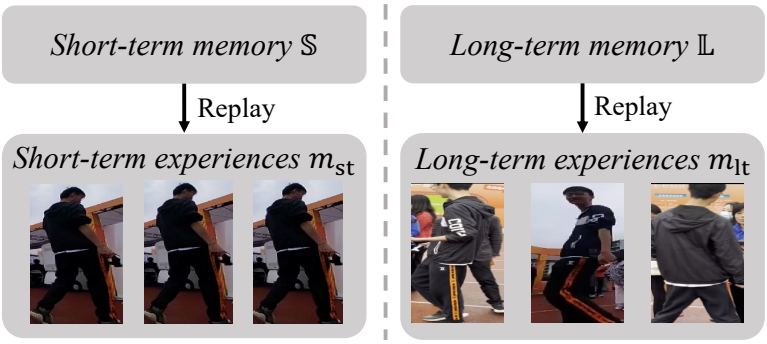


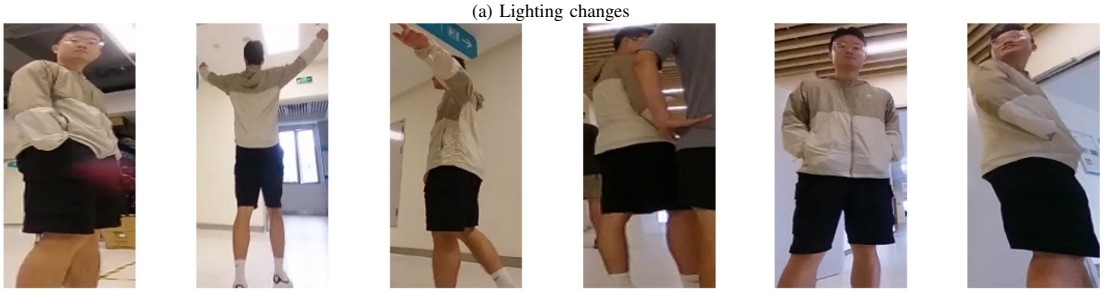

Fig. 7. Both short-term and long-term experiences ($m_{\mathrm{st}} \cup m_{\mathrm{lt}}$) are responsible for training the feature extractor with $m_{\mathrm{st}}$ and $m_{\mathrm{lt}}$ being sampled from short-term memory $\mathbb{S}$ and long-term memory $\mathbb{L}$, respectively. $\mathbb{L}$ contains sparse yet valuable historical samples, maintained by the *memory manager*. Besides, the target classifier is trained with $m_{\mathrm{st}}$ sampled from $\mathbb{S}$, which stores the most recently observed samples, representing the latest knowledge.

### C. Visual Examples during RPF Task

Sampled images are shown in Fig. 8 consisting of the observation of the target person during the robot-person-following task. We can observe significant changes in the appearance of the target person from left to right including lighting and viewpoint changes. In such situations, previous methods relying short-term experiences would fail to re-identify the target person after occlusion. For example, as shown in Fig. 8 (b), short-term experiences capture the latest observation, i.e., the back view of the target person. Consequently, when the target person reappears with a front view, these short-term experiences fail to re-identify them.

In contrast, our method utilizes both short-term and long-term experiences. Specifically, we fine-tune the feature extractor within our OCL-assisted RPF-ReID module using both types of experiences. This approach constructs a complete representation of the target person, considering long-term experiences (e.g., the target person's front view) and short-term experiences (e.g., the target person's back view). As a result, our method can re-identify the target person even when they reappear with a front view.

(a) Lighting changes



(b) Viewpoint changes

Fig. 8. Sampled images of the target person during the robot-person-following task. We can observe significant changes in the appearance of the target person from left to right. (a) Lighting changes: bright light in the corridor, dim light in a corner, and bright light again in the elevator. (b) Viewpoint changes: the person's front view, back view, occlusion, and front view again.

### *D. Target-ReID Lifecycle*

**Algorithm 1:** Target-ReID Lifecycle

**Input:** Current image $\mathbf{I}$ and tracked people $\{\mathbf{B}, \mathbf{p}\}_i$ representing bounding boxes and positions, target person's identity $id$, target confidence $s$, short-term memory $\mathbb{S}$, long-term memory $\mathbb{L}$, feature extractor $f$ and target classifier $g$

**Output:** Target person's position $\{\mathbf{p}\}_{id}$ in the current frame

Extract image patches $\mathbf{M}$ from $\mathbf{I}$ and $\mathbf{B}$;
Construct the observation set $\{\mathbf{M}, \mathbf{y}\}_i$ where $\mathbf{y} = 1$ if $i == id$, otherwise $\mathbf{y} = 0$;
Extract features $\mathbf{F}$ from $\mathbf{M}$ with $f$;
**if** $id \in \{i\}$ **then**
  Estimate $s$ of the target person based on Eq. 3;
  **if** $s > \delta_{\text{sw}}$ **then**
    Consider $\{\bar{i}\}$ as identities of negative tracks;
    $\{\mathbf{M}, \mathbf{y}\}_{id} \rightarrow \mathbb{S}$, $\{\mathbf{M}, \mathbf{y}\}_{\bar{i}} \rightarrow \mathbb{S}$ based on FILO rule;
    Sample $m_{\text{st}}$ from $\mathbb{S}$;
    Train $g$ with $m_{\text{st}}$ based on Eq. 4;
    ### Separate Thread ###
    $\{\mathbf{M}, \mathbf{y}\}_{\bar{i}} \rightarrow \mathbb{L}$ based on FILO rule;
    $\{\mathbf{M}, \mathbf{y}\}_{id} \rightarrow \mathbb{L}$ if it is a keyframe based on Eq. 5;
    Consolidate $\mathbb{L}$ with OCL techniques if $\mathbb{L}$ is full;
    Sample $m_{\text{lt}}$ from $\mathbb{L}$;
    Train $f$ with $m_{\text{st}}$ and $m_{\text{lt}}$ based on Eq. 1;
    ### Separate Thread ###
    **Return** target position $\{\mathbf{p}\}_{id}$;
  **else**
    Let $id = -1$, indicates id switch between the target person and other people;
    **Return** $\emptyset$;
**else**
  Estimate $s$ of the $i_{th}$ person based on Eq. 3;
  **if** $s > \delta_{\text{reid}}$ for consecutive $\zeta_{\text{reid}}$ frames **then**
    Let $id = i$, indicates successful target person ReID;
    **Return** target person's position $\{\mathbf{p}\}_{id}$;
  **else**
    **Return** $\emptyset$;